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MILITARY SPECIFICATION

TRUCK, TANK A/S32A-2

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense."

1. SCOPE

1.1 Scope. This specification covers a 2,600 gallon tank truck capable of handling individually demineralized water, JP-4 or JP-5 turbine fuel conforming to MIL-T-5624, or aviation gasoline conforming to MIL-G-5572.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, stanards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

KKK-T-2109 Trucks and Truck Tractors: Commercial, Diesel and Gasoline Engine Driven

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMEDTB, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2320

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Military

MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-B-5087	Bonding, Electrical, and Lighting Protection, for Aerospace Systems
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-M-6404	Meter, Volumetric, Positive Displacement, Liquid, Type F-1, Aircraft Fuel, 300 Gallons Per Minute
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-10215	Speedometers, Tachometers, Flexible Shaft Assemblies, Flexible Shaft Assembly Cores, Adapters, and Transmitters
MIL-C-10387	Coupling, Clamp, Pipe, With Bolts and Synthetic Rubber Gasket, for Grooved Pipe and Tube
MIL-H-26521	Hose Assembly, Rubber, Fuel, Collapsible, Low Temperature, With Reattachable Couplings
MIL-C-27487	Coupling Halves, Quick Disconnect-Cam-Locking Type
MIL-H-38209	Hose Reel Assembly, Fuel Servicing, General Requirements For
MIL-C-38404	Coupling, Hose, Reattachable Screw-on
MIL-M-43719	Marking Materials and Markers, Adhesive, Elastomeric, Pigmented;
MIL-R-83232	Reel, Static Discharge, Grounding
MIL-W-83420	Wire Rope, Flexible for Aircraft Control
MIL-F-83428	Filter Element, Fluid, Pressure MXU-649E
MIL-N-87963	Nozzle, Fuel, Aircraft, Overwing, Type MD-3

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-12	Abbreviations for Use on Drawings, and in Specifications, Standards and Technical Documents
MIL-STD-100	Engineering Drawings Practices
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-276	Impregnation of Porous Nonferrous Metal Castings
MIL-STD-461	Electromagnetic Emission and susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-889	Dissimilar Metals
MIL-STD-1223	Administrative Wheeled Vehicles Treatment, Painting, Rust-proofing, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

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MIL-STD-1595	Aerospace Welder Performance Qualification
MS3506	Connector, Receptacle, External Electric Power, Aircraft, 28 Volt, DC Operating Power
MS25083	Jumper Assembly, Electric, Bonding and Current Return
MS51118	Pintle Assembly, Towing, 40,000 lbs Capacity, Manual Release

2.1.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION

Federal Motor Carrier Safety Regulations.

DEPARTMENT OF AGRICULTURE

Forest Service Standard 5100-1

ENVIRONMENTAL PROTECTION AGENCY

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services

Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

ANSI H35.1	Wrought Aluminum
ANSI S1.4	General Purpose Sound Level Meters (1971 Issue)

(Application for copies should be addressed to the American National Standards Institute, 10 East 40th Street, New York 10016).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Section VIII	Pressure Vessels
Section IX	Welding Qualifications

(Application for copies should be addressed to the American Society of Mechan-

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ical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017).

SOCIETY OF AUTOMOTIVE ENGINEERS, INCORPORATED

SAE Recommended Practices

J336	Sound Level for Truck Cab Interior
J350	Engine, Medium Size, Spark Arrester Test Procedure
J366a	Exterior Sound Level for Heavy Trucks and Buses
J534	Lubrication Fittings
J683	Tire Chain Clearance - Trucks, Buses, and Combination of Vehicles

(Application for copies should be addressed to the Society of Automotive Engineers, Inc.; 400 Commonwealth Drive, Warrendale, PA 15096).

AMERICAN WELDING SOCIETY

B3.0 Standard Qualification Procedure

(Application for copies should be addressed to the American Welding Society, United Engineering Center, 345 East 47th Street, New York, New York 10017).

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 407

(Applications for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston MA 02110).

STATE OF CALIFORNIA

Vehicle Code of California

(Application for copies should be addressed to Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, and specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4).

3.2 Components. This truck shall consist of the following major components:

Item or System Name	See Requirement
Chassis	3.22.1
Tank	3.22.2

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Pumping and filtration compartments 3.22.3
Pumping system 3.22.4

3.3 Dimensions. The overall dimensions shall be reduced based upon the exact configuration of the truck to be furnished and the air transportability requirements specified herein.

3.3.1 Width. The overall width of the truck shall not exceed 96 inches with all access openings closed.

3.3.2 Height. The overall height of the truck shall not exceed 105 inches with all access openings closed.

3.3.3 Length. The overall length of the truck shall not exceed 31 feet.

3.3.4 Dimensional tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the truck, they shall be held or limited accordingly.

3.4 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.5 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced document shall be suitable in every respect for this type equipment. Except as otherwise specified, wood shall not be used.

3.5.1 Protective treatment. When materials are used in the construction of the truck that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings that will crack, chip, or scale with use, age, or extremes of climatic and environmental conditions shall not be used.

3.5.2 Fungusproof materials. Materials which are not nutrients for fungi shall be used whenever possible. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent acceptable to the procuring activity.

3.5.3 Bonding. To insure that all parts of the tank, pumping system, and truck chassis are at the same (ground) potential, all system components shall be bonded together. Whenever the electrical continuity of the system is interrupted by a nonmetallic insulating material or coating at the contacting surfaces, a bonding strap or jumper in accordance with the table of MIL-B-5087 entitled Electrical Bond Classes of Application and MS 25083 shall be installed.

3.5.4 Metals. Unless otherwise specified herein, the tank and pumping system and all components which come into contact with the fluids specified in 3.6.2 shall be fabricated of aluminum or 300 series stainless steel. All aluminum parts except aluminum castings shall be fabricated of aluminum alloys of the 1000, 3000, 5000, or 6000 series in accordance with ANSI H35.1. Aluminum casting alloys shall be limited to types B214 and 356. Other aluminum casting

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alloys and stainless steels may be substituted upon approval of the procuring activity.

3.5.4.1 Impregnation of castings. Aluminum castings may be impregnated in accordance with MIL-STD-276 to prevent minor weeping. Alternates to MIL-STD-276 may be used if approved by the procuring activity.

3.5.4.2 Dissimilar metals. Unless protected against eletrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5.5 Elastomers. Elastomeric materials used in the tank and pumping system which come into contact with fluid being transported and pumped shall be compatible with aviation turbine fuels, aviation gasoline, tap water, and demineralized water.

3.6 Design and construction. The truck shall be designed to store and transport water, turbine fuel or aviation gasoline.

3.6.1 Construction. The vehicle shall be constructed so that parts do not work loose in service, and shall withstand the loads, shocks, vibrations and other conditions incident to operation, shipping and storage with minimum loss of time for maintenance, repair and servicing. The design of the vehicle shall ensure safe, efficient and economical operation. All assemblies, controls and installed equipment shall be located so that there is no adverse interference with each other or with the operation, and shall be readily accessible for maintenance, operation, removal and replacement. There shall be no evidence of accelerated wear, failure or permanent deformation when operated as intended.

3.6.2 Functional design. The truck shall be designed to satisfactorily receive, transport, store, and service demineralized water having a total solid content of less than 5 parts per million. The water shall not result in the corrosion of any part of the truck. The dissolved solid contaminant content of the water shall not be appreciably affected by storage and handling in the truck. The truck shall be designed to receive, transport, store, and service fuel in accordance with MIL-G-5572 and MIL-T-5624 by deactivation of the heater electrical system or substitution of piping for the heater, the addition of flowmeters, and conversion of the filter.

3.6.2.1 World-wide operation. The design of the truck shall allow world-wide operation at fixed bases.

3.6.3 Fastening devices. All screws, bolts, nuts, pins and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except on the commercial chassis. Topped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter.

3.6.4 Reliability. The truck, including all components, shall have a Meantime Between Failure (MTBF) (see 6.3.2 and 6.3.2.1) of not less than 100 hours demonstrated at 0.9 confidence.

3.6.5 Maintainabilty. The vehicle shall be designed and constructed to provide:

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a. Parts which are consistent with the performance required herein.

b. A minimum time requirement for assembling, dissassembling, locating trouble sources and maintaining the vehicle. Where practical, components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing and replacing parts and components.

d. Use of readily available standard tools and equipment for maintenance.

3.6.5.1 Maintenance downtime. The truck shall be designed and constructed to permit maintenance by military maintenance technicians in not more than the downtimes specified in 3.6.5.2 and 3.6.5.3. Not more than two technicians shall be required simultaneously for any task.

3.6.5.2 Corrective maintenance downtime. The mean corrective maintenance downtime for all levels of maintenance below depot or equivalent level shall be not more than 30 minutes with a related maximum corrective maintenance downtime of not more than 1-1/2 hours.

3.6.5.3 Maximum preventive maintenance downtime. Maximum preventive maintenance downtime, excluding depot or equivalent level maintenance and major overhaul, shall be not more than 45 minutes.

3.6.6 Component accessibility. The truck shall be designed to enable ease of component removal, servicing, general maintenance and repair under various environmental conditions. Any design which renders routine maintenance unduly difficult under normal field operations will not be acceptable.

3.6.7 Handtools. The truck shall be designed and constructed in a manner that will allow general maintenance, assembly, and disassembly to be performed to the maximum practical extent using common handtools.

3.6.8 Welders and welding.

3.6.8.1 Welders. All welders employed in the fabrication of the vehicle shall be certified in accordance with the applicable requirements of MIL-STD-1595, The American Welding Society, or The American Society of Mechanical Engineers.

3.6.8.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof and service loadings.

3.6.8.3 Weld Beads. Weld beads shall be thoroughly cleaned, and spattering shall be removed from the tank.

3.6.9 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where use of high lubrication pressure may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication, or which are likely to be overlooked because of inaccessibility.

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3.6.10 Air transportability. The vehicle shall be air transportable in accordance with MIL-A-8421 in Types C-130, C-141 and C-5 Aircraft. Removal of top mounted equipment shall be permitted, if necessary, to provide the required height for air transport. Removal and installation of equipment shall be limited to removal and attachment of standard fasteners and fittings with common hand tools. The weight of individual parts or assemblies which must be handled or positioned shall not exceed the limits specified in MIL-STD-1472 for a one person lift. All equipment removed shall be stored on the vehicle. All necessary caps and plugs shall be provided to permit driving and storage of the vehicle while in the air transport configuration. Removal or reinstallation of the equipment shall take no longer than 15 minutes for two technicians using common hand tools.

3.6.10.1 Tiedowns. The truck shall be provided with sufficient tiedown points so located that the vehicle can be restrained in all directions during air transport. Tiedowns shall be designed for vehicle restraint against forces of 4.5g forward and backward, 2g vertical and 1.5g lateral. The attachment points of the truck shall be designed for use with attachment points in the aircraft floor. These aircraft attachment points have 10,000 pounds capacity in a 20 inch grid pattern. The 25,000 pound attachment points have approximate 40 inch centers along both sides of the aircraft floor. The tiedown points on the truck shall be designed and located on the basis that restraining chains will be at an angle of 30° with the aircraft floor. Vehicle attachment point loading capacities shall be downgraded accordingly from the standpoint of restraining horizontal, vertical and lateral g-loads on the truck.

3.6.10.2 Clearance. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure on the truck. The tiedown points shall have a minimum clear opening of 4 inches to allow attachment of a chain tiedown with a grabhook.

3.7 Safety. As a minimum, the contractor shall comply with the provisions of the paragraphs of MIL-STD-882 entitled System Safety Criteria and Considerations, Hazard Levels, System Safety Precedence, Design Criteria/Specifications.

3.8 Weight distribution. The weight distribution of the vehicle in any loading condition shall not be unbalanced or improperly distributed. The difference in loading between the left and right wheels of any axle shall not exceed 2.5% of the total load on the axle.

3.9 Electromagnetic Interference. The vehicle shall comply with the requirements of the MIL-STD-461, Class C1, Group II for electromagnetic interference, except that the limits for UM03 (Figure 8-1) shall be relaxed by 20db.

3.10 Foolproofness. Where improper installation of an item could cause a malfunction, an unsymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

3.11 Servicing provisions. Drains, grease fittings, lines and check points for all components shall be located so that they are readily accessible and do not require special tools for proper servicing. Where winterization is specified, all hatches and releases shall be such that an arctic clad person can gain access to all service points and perform necessary maintenance.

3.12 Component protection. All space in which work is performed during

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operation, service and maintenance of the vehicle shall be free of hazardous protrusions, sharp edges or other features which may cause injury to pesonnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, shall be guarded or insulated to the extent necessary to eliminate the hazard to personnel.

3.13 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the vehicle.

3.14 Bushings and bearings. Bushings or bearings, as appropriate, shall be provided at any location where there is relative rotation between two parts, except hinges.

3.15 Markings. Unless otherwise specified, markings for vehicles painted forest green shall be applied with black paint, Color Number 37038 of FED-STD-595, and markings for vechicles painted desert sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.15.1 Identification markings. Unless otherwise specified (see 6.2), the vehicle shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.15.2 Operational markings.

3.15.2.1 Lettering and numbering. Unless specified (see 6.2), all lettering and markings shall be stenciled as follows:

a. "DEMINERALIZED WATER" shall be stenciled in 4 inch lusterless red paint, Color Number 31136, centered vertically and horizontally on the left and right side of the cargo tank, and on the rear of the tank 8 inches from the top.

b. "CAUTION - WATER NOT FIT FOR DRINKING" shall be stenciled in 4 inch lusterless red letters directly below "DEMINERALIZED WATER."

3.15.2.2 Data plates. Instructions and identification data shall be permanently and legibly etched upon metal plates in accordance with MIL-P-514. The instruction plates shall be permanently attached and located on the inside of the compartment doors so as to be most accessible and visible to the operator standing in front of the control panel. The instruction and other data plates shall be aluminum and not less than 0.050 inch thick. Component identification plates shall be aluminum and not less than 0.032 inch thick. All plates shall be anodized or etched to produce raised natural metallic colored markings with a dark color background. All data plates, placards, charts, instruction plates, shall conform to MIL-P-514, and be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.15.2.2.1 Nameplates. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130. The plates shall be made of a noncorrosive steel. In addition, the acceptance date and gross weight shall be included.

3.15.2.2.2 Control markings. All controls, valves, gages and indicators

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used in the operation of the vehicle shall be identified by securely attached name-plates of such construction that exposure to fuel, oil, water, dirt, light, etc., will neither fade nor eradicate them. Tags or decals shall not be used.

3.15.2.2.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise and adequate to enable operation of the vehicle without damage to equipment or injury to personnel, and shall refer to the components as identified by nameplates.

3.15.2.2.4 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and components which require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

3.15.2.2.5 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

a. Side and rear silhouette views of the vehicle.

b. Horizontal and vertical location of the center-of-gravity of the vehicle in transport configuration.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width and height.

f. Tiedown instructions.

3.15.2.2.6 Hose reel instruction plate. A hose reel instruction plate shall be provided which cautions the operator to control the rewind rate of the hose onto the hose reel by holding the nozzle, as damage could occur to the hose reel and nozzle if the hose is allowed to freely rewind onto the reel. The instruction plate shall be mounted in a conspicuous location near the hose reel.

3.15.2.2.7 Heater instruction plate. A concise, step-by-step operating instructions plate containing procedures to be followed for starting, operating and stopping the heater, and precautionary measures to be taken by the operator, shall be permanently attached to the heater control panel.

3.15.2.2.8 Identification plates. Valves, switches, levers and controls used in the operation of the truck shall be properly identified by identification plates. The data plates shall incorporate a numerical or alphabetical system of identification and an indication of the relative position of the lever or control handles such as; on, off, 12V, etc.

3.15.3 Marking of tiedown points. The location and capacity of each tiedown point specified in 3.6.10.1 shall be clearly marked. The letters and numbers of the markings shall be 1/4 inch high.

3.15.4 Marking of control lever. The control lever specified in 3.22.1.6.1

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shall be permanently marked to indicate the pump and drive positions.

3.15.5 Marking of no-combustion light. The red no-combustion light to be located over the heat switch as specified in 3.22.4.5.8.4 shall be suitably marked.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. In addition, the acceptance date and gross weight shall be included.

3.17 Screw assemblies. Unless torque is specified, threaded fasteners shall be tight, and shall not work loose during testing or service usage. All threads shall match the mating threads and be free of burrs.

3.18 Riveting. Riveting operations shall be performed to ensure tight fit and complete heading of the rivets.

3.19 Gear and lever systems. Gear and lever systems shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, and other irregularities. Where required for accurate adjustments, gear assemblies shall be free of backlash.

3.20 Cleaning. The truck and particularly the tank assembly shall be thoroughly cleaned. Loose, spattered or excess solder, welding slag, stray bolts and nuts, pipe compound, rust, metal particles, and other foreign matter shall be removed during and after final assembly.

3.21 Performance.

3.21.1 Environmental conditions. The vehicle shall be capable of satisfactory storage, start and operation under the following environmental conditions with truck stabilized:

a. Temperature ranging from -40° F to 125° F.

b. Exposure to relative humidities up to 100 percent.

c. Part-time (five percent of the time) exposure to salt fog, five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0ml/80 cm^2/hour.

d. Part-time (five percent of the time) exposure to blowing dust; velocity 1750 ft/min, concentration 0.5 g/ft^3. Part-time (five percent of the time) exposure to blowing sand, velocity 5700 ft/min, concentration 0.0773 g/ft^3.

e. Wind velocities of up to, and including, 70 mph.

3.21.2 Pumpings system operation. The pumping system shall provide for any servicing rate from 0 to 130 gpm by control of the overwing nozzle. This system shall be capable of circulating and heating water from the tank at a flow rate of at least 100 gpm, and shall exhibit no instability in pressure or flow rate under any operating conditions. The pumping system shall have no electrical equipment or devices associated with, or incorporated as, a part of the pumping system except as specifically required herein, and shall be capable of providing performance as specified below.

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a. Pump from the tank through the filter and heater and out through the servicing hose at any flow rate from 0 to 130 gpm. Once the pumping system has been started, flow shall be regulated on demand by control of the overwing nozzle.

b. Pump from the tank through the filter and heater and bypass return to the tank at a flow rate of at least 100 gpm. This pumping mode shall be used for heating the contents of the tank.

c. Pump from the suction stub through the filter and heater and bypass return to the tank at a flow rate of at least 100 gpm. This performance shall be attained with a 50 foot, 2 inch ID noncollapsible hose attached to the suction stub.

d. Evacuation of 90% of the product in the fully flooded dispensing hoses within 1 minute.

3.21.2.1 Closed discharge. The pump shall operate against a closed discharge for periods of 30 minutes without appreciable wear, damage or overheating. The closed discharge condition shall be attained by operating the unit at maximum rated throttle setting with all manual valves closed, except the tank shutoff valve and pump shutoff valve, if provided, and the system bypass. The water temperature in the pump shall not exceed 160° F during the 30 minutes period with an ambient temperature and tank water temperature of approximately 70° F. This performance shall be attained with a maximum of 1,000 gallons of water in the tank.

3.21.2.2 Dry operation. The pump shall operate at maximum rated pump speed for 5 minutes without appreciable wear or damage after the unit has been completely drained.

3.21.2.3 Suction lift. The pump shall be capable of a suction lift of 15 inches Hg.

3.21.3 Tank.

3.21.3.1 Static pressure. The tank shall withstand an internal static pressure of at least 3 psi without evidence of leakage.

3.21.3.2 Hydrodynamic loading. The tank shall be capable of being filled at a flow rate of 600 gpm through a 4 inch pipe in the tank manhole without damage.

3.21.3.3 Hydrostatic pressure. The piping system and all components, except the MD-3 servicing nozzle, specified in 3.22.4.9, shall withstand a hydrostatic pressure of 1 1/2 times the maximum pressure encountered during operation of the unit or minimum of 180 psi, whichever is greater. The servicing nozzle shall withstand the hydrostatic test pressures required by MIL-N-87963.

3.22 Details of Components.

3.22.1 Chassis. The chassis shall be the manufacturer's standard, except as modified to comply with requirements specified herein. The chassis shall be 6 X 4, and shall conform to KKK-T-2109, Type I, Class C, and also as specified herein.

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3.22.1.1 Cooling system. The chassis manufacturer's heavy duty cooling system shall be provided.

3.22.1.2 Air pollution control. All vehicles shall comply with the Environmental Protection Agency Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture.

3.22.1.3 Alternator. The alternator shall be of sufficient capacity to maintain an adequate supply of electrical power under all operating conditions with all 12V accessory components of the truck operating simultaneously. The alternator shall be of not less than 80 amp capacity.

3.22.1.4 Fuel tank(s). A fuel tank(s) of the chassis manufacturer's current design and construction shall be furnished and mounted outside of the pumping compartment. The tank(s) shall have a minimum capacity of 100 gallons, and shall conform to current Motor Carrier Safety Regulations.

3.22.1.5 Transmission. The truck shall be equipped with an automatic or semi-automatic transmission in accordance with KKK-T-2109.

3.22.1.5.1 Selection of ranges of gears. The transmission shall provide for driver selection of the ranges of gears required for efficient operation throughout all speeds to be encountered during operation of the truck. The shift control shall incorporate a positive indicator for each gear range selected. The transmission shall be so protected that the operator cannot inadvertantly shift the truck into reverse.

3.22.1.5.1.1 Safety devices. A safety device shall be installed to prevent starting the engine in any setting other than neutral.

3.22.1.5.2 Braking. The transmission shall permit effective braking by the engine when the truck is decelerating or descending grades.

3.22.1.6 Power takeoff. A split shaft power takeoff shall be furnished. The power takeoff shall have sufficient capacity to drive the dispensing pump and 24V alternator for the maximum load conditions to be encountered during operation of the truck. The power takeoff and pump gearbox gear ratios shall be selected to provide for pump operations at sufficient engine speeds to assure adequate engine and gearbox lubrication and engine cooling.

3.22.1.6.1 Power Takeoff Control (PTO). The power takeoff shall be air operated and controlled by the driver while sitting in the driver's seat. The control lever shall not interfere with normal operation or access to the driver's seat by the driver entering or existing the cab. Other controls, such as a driveshaft brake, shall be installed as required to provide a positive, smooth mechanical engagement. Any engagement device which can cause clashing of mechanical parts or heavy shock loading will not be acceptable. An operational data plate shall be installed in the truck cab next to the PTO control giving step-by-step instructions for engaging, operating and disengaging the PTO.

3.22.1.6.2 Power takeoff safety devices. An interlock system shall be provided to prevent simultaneous propelling of the truck and operation of the pump and 24V alternator. An interlock system shall be provided to prevent shifting

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of the power takeoff control in either direction unless the engine is at idle speed and the transmission is in neutral. A safety device shall be provided to prevent operation of the auxiliary throttle when the power takeoff is in the vehicle drive position. A positive indicator device shall be installed to show when PTO is in pump mode. Indicator shall be mechanically actuated by PTO shifter shaft.

3.22.1.7 Tires and wheels. Tires shall be standard commercial type tires. Standard commercial heavy duty type wheels shall meet the Tire and Rim Association requirements specified for this type vehicle. Unless specified (see 6.2), a spare tire, wheel, carrier and tools are not required.

3.22.1.8 Brakes. Brakes shall be installed on all wheels. The service brakes shall stop the truck in any load condition within 35 feet from a speed of 20 mph on dry, hard, approximately level road surfaces free from loose material. The parking brakes shall hold the truck in any load condition in either direction on a slope of at least 20°. Spring applied parking brakes shall be installed on both rear axles (four wheels).

3.22.1.9 Cab dimensions. The distance from the centerline of the front axle to the rear of the cab shall not exceed 40 inches. The maximum unloaded height from the ground to the top of the cab or to any permanent member of the cab shall not exceed 105 inches. The overall height of the cab shall be further reduced as required based upon the exact configuration of the truck to be furnished and the air transportability requirements specified herein.

3.22.1.10 Pintle hook. A pintle hook conforming to MS 51118 shall be mounted on the rear of the truck. The pintle hook shall be mounted to permit a 25,000 pound drawbar pull without causing permanent distortion or damage to the mounting structure. The procuring activity will specify whether or not pintle hooks are to be provided on production units (see 6.2). All prototype and first article trucks shall include installed pintle hooks, and all production trucks shall include complete provisions for installation of pintle hooks by field activities, if not included on the production trucks.

3.22.1.11 Bogie axle spacing. The bogie shall have an axle spacing of more than 48 inches, but not more than 54 inches.

3.22.1.12 Rear bumper. A full width, heavy duty rear bumper shall be mounted on the chassis frame. The bumper shall be constructed of a steel channel section. The outer corners of the bumper shall be rounded and shall incorporate other protection, as required, to minimize the hazard to personnel walking around the truck. This bumper shall be designed to allow pushing of the fully loaded vehicle without damage, and shall have appropriate cutouts for recess mounting of the brake, parking and directional signal lights.

3.22.1.13 Exhaust system. Exhaust system shall conform to Vehicle Motor Carrier Safety Regulation 393.83. Exhaust tailpipe shall extend to the rear of the vehicle, or at least 10 inches above the cab and be so located as to minimize the possibility of exhaust fumes entering the body of the vehicle. The exhaust system shall be provided with a spark arrester comforming to Forest Service Standard 5100-1 and SAE J350 for at least 80% efficiency. The spark arrester shall be provided with a means for cleaning out trapped particles. The muffler shall be so positioned and mounted that it cannot be easily damaged.

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3.22.1.14 Lighting and wiring systems. The truck lights and reflectors shall conform to, and be installed, in accordance with current Motor Carrier Safety Regulations and as specified herein. Wiring, lamps and all electrical equipment and components installed in the hazardous vapor area shall conform to the following requirements; except the hose reel electrical components, which shall be in accordance with MIL-H-38209, and the heater.

3.22.1.14.1 Wiring. Wiring for lamps shall be no smaller than 12 gage, stranded, with thermoplastic insulation, with the exception of lamp pigtails, which may be 16 gage, but no longer than 6 inches. Wiring shall conform to SAE Standards for low tension, insulated cable. All wiring shall be identified by coloring, numbering or a combination of both.

3.22.1.14.1.1 Wiring system installation. The wiring system shall be explosion proof (vapor proof) in accordance with current flammable tank vehicle industry practice. All wiring shall be enclosed in rigid, metal conduit firmly attached to the truck. All switches and circuit protection devices installed in the hazardous vapor area shall be of permanent vapor proof design, or enclosed in vapor proof boxes to prevent vapor from entering the arcing device. Wiring connections to these devices shall be through the metal conduit threaded to the components and shall be designed to withstand a 3 psi positive or negative pressure without leaking. The external power receptacle shall be in accordance with MS3506.

3.22.1.14.1.2 Protection devices. Overload circuit breakers shall be used in lieu of replaceable fuse type devices. All circuits shall be protected by circuit breakers. Protective devices shall be connected and identified in an organized manner to facilitate determination of malfunction.

3.22.1.14.2 Lamps. All lamps installed in the hazardous vapor area shall be of permanent vapor proof and waterproof design. The lamp body shall be nonferrous and shall incorporate a screw type fitting for making a vapor proof connection with the wiring conduit. The lens shall be heavy duty and shall withstand normal removal and installation without cracking or breaking. The lens shall be designed to provide a vapor proof seal with the lamp body. The lamp assembly shall be designed to withstand a 3 psi positive or negative pressure without leaking. Cements, sealant, or similar material shall not be used to attain the vapor proof seal.

3.22.1.14.2.1 Lamp bulbs. The lamp bulbs shall be of the rough service type.

3.22.1.15 Battery. Battery capacity shall be sufficient to meet the low temperature starting and operating condition specified herein. The battery shall be so installed that the electrolyte can be easily checked without removing the battery. The battery compartment shall be vented. The battery shall be of the maintenance free type.

3.22.1.16 Fenders. Fenders shall be provided over the rear wheels of the truck. The fenders shall be of heavy gage steel and supported to withstand a static load of 400 pounds placed on top of the fenders. The outer edges of the fenders shall be rolled or reinforced with rolled steel angle. The underside of the fenders shall be given a heavy application of high grade vehicle undercoating. Adequate clearance shall be maintained for easy installation of tire chains and shall follow the guide set up in SAE J683.

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3.22.1.16.1 Skirting. Skirting may be supplied in lieu of fenders provided, it is designed and constructed to provide the required protection.

3.22.1.17 Mudflaps. Substantial flexible mudflaps shall be mounted behind the front wheels, the bogie and in front of the bogie. The ground clearance of the mudflaps shall be not more than 1/3 of the distance from the center of the closest axle to the center of the flaps under any load condition. The mudflaps shall be mounted within 12 inches of the tires, and shall be at least as wide as the tires. The flaps in front of the bogie shall be supported so as not to interfere with the running gear. If the truck is so constructed that these requirements are accomplished by means of fenders or body construction, mudflaps are not required.

3.22.1.18 Driveshafts. The driveshafts, including the power takeoff, shall be protected in a manner similar to that specified for buses by Motor Carrier Safety Regulation 393.89 to prevent damage to the tank and fluid carrying components.

3.22.1.19 Rear springs. The rear springs shall be adjusted such that when the tank is half filled with demineralized water, the truck frame is level plus or minus $1^{O}$when the truck is on a level surface.

3.22.2 Tank. The tank shall be of the fully insulated, single compartment type, with a 2,600 gallon liquid capacity plus at least 3% expansion space. The tank shall be constructed of 5000 series sheet aluminum of sufficient thickness to provide complete structural integrity of the tank for the life of the truck. The minimum sheet of the tank shall be 0.160 inch. The tank shall be elliptical or modified rectangular cross section. The tank interior shall have a bright finish. Water stained aluminum will not be permitted.

3.22.2.1 Sump. A sump shall be incorporated at the lowest point of the tank. The sump shall have a minimum capacity of 8 gallons to allow accumulation of solid cotaminants and water when the truck is used for fuel servicing. The tank shall be so designed that water will drain into the sump and will neither be trapped by the baffles, nor stand in pockets when the truck is on flat, horizontal pavement.

3.22.2.1.1 Sump drain. A 2 inch drain valve shall be installed in the sump to permit complete drainage of the tank and sump. The drain valve shall be protected against flying objects, and shall be easily accessible. The outlet of the drain shall be directed toward the ground.

3.22.2.2 Baffle and heads. The baffle and exterior heads shall be designed, constructed and installed in accordance with current Motor Carrier Safety Regulations for flammable liquid tanks. The baffle shall be designed and constructed to permit inspection and cleaning of the entire tank. Ring stiffeners or other reinforcement designs shall not be substituted for the baffle. A 175 pound person shall be capable of entering any section of the tank through the baffle. Baffles shall be flanged and dished. A half round or full round opening of not less than 3 3/8 inch radius shall be provided at the bottom of the baffles. Openings of equivalent size shall be provided at the top of the baffles. The cross sectional area of each baffle shall be not less than 80% of the cross sectional area of the tank. The edges of the half round or fullround openings shall have a flange of not less than 3/4 inch.

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3.22.2.3 Insulation. The tank shall be completely insulated with a foam type material. The material shall not deteriorate in water or fuel. The material shall be cemented to the tank surface in order to prevent packing and settling. Heat loss through the total insulation thickness provided shall not exceed 0.2 BTU per hour per square foot per° F temperature differential.

3.22.2.3.1 Sheet metal covering. The insulating material shall be fully enclosed with a sheet metal covering which will form the external covering of the tank. The sheet metal covering shall be designed and constructed to prevent the entrance of water and fuel. Careful attention shall be given to the design and construction of the covering to prevent the entrance of water spilled around the manhole during loading, and around the tank mounting due to road splash. The sheet metal covering shall be aluminum. The covering shall incorporate reinforcement as required to provide a rigidly constructed structure.The entire top surface of the tank and cover shall be capable of supporting two men, at 220 pounds each, standing at any point without permanent deformation of any portion of the tank and cover. The sheet metal cover shall not be joined together with hollow rivets.

3.22.2.4 Cleaning stubs. The tank shall incorporate 4 inch cleaning stubs and caps in the top of each tank section, as separated by the baffles, except for the section incorporating the manhole. Each end section cleaning stub cap shall incorporate a fusible plug.

3.22.2.5 Recirculation manifold. The pumping system line shall terminate in a recirculation manifold installed in the tank. The recirculation manifold shall be designed to provide uniform mixing of the water during the heating system operation. The recirculation manifold shall provide for approximately uniform flow of water in each section of the tank as separated by the baffles.

3.22.2.6 Tank calibration marker. A fixed marker that is visible through the manhole shall be provided inside the tank to indicate when the tank has been filled with 1,000, 2,000 and 2,600 gallons of water.

3.22.2.7 Manhole. The tank shall have a circular manhole with a minimum diameter of 20 inches. The manhole shall have a 10 inch, watertight, hinged filler cover with necessary safety features to eliminate the possibility of a spark occuring during opening or closing of the cover. The manhole shall be equipped with a Buna N, or equivalent, gasket to insure a watertight seal between the collar and the cover. The entire top of the manhole shall be removable for entrance into the tank by removing the bolt-on clamping ring.

3.22.2.7.1 Fusible plugs. Renewable type fusible plugs with firescreens shall be installed in each end tank section cleaning stub cap.

3.22.2.8 Vents. The tank shall be equipped with at least two identical removable vents installed in the manhole cover. Each vent shall be designed and adequately sized to allow water to be pumped into and out of the tank at 0 to 300 gpm with the manhole hinged filler cover closed. Each vent shall not open until the pressure or vacuum in the tank reaches 0.2 psig and shall not permit the pressure in the tank to exceed 2 psig at 300 gpm flow rate, nor the vacuum to exceed 1 psig at 300 gpm flow rate. The vent shall remain closed at all other times. The vents shall be so designed that malfunctioning and jamming will not occur. Metal alloys which could corrode and cause malfunction shall not be used.

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3.22.2.9 Catwalk. A catwalk shall be provided, and shall be integral with, and securely attached to the tank. The catwalk shall be constructed of embossed, diamond type, expanded or equivalent metal safety tread. In the event the expanded or equivalent metal is attached to the tank insulation cover, sufficient reinforcement shall be incorporated so that deformation will not occur under a load of 180 pounds over any 16 square inches of the catwalk or a total load of 600 pounds on any 3 square feet of the catwalk. The catwalk shall be at least 30 inches wide, and shall span the manhole, extend the length of the tank and incorporate an adequate landing platform to accomodate the boarding ladder specified in 3.22.2.10. Means shall be provided for draining entrapped water from the catwalk. The catwalk shall be so designed that liquid spilled, while filling the tank through the manhole, will drain off the rear of the tank. Appropriate provisions shall be incorporated to permit cleaning to assure that leaves, etc., are not restricting the flow of the drainage provisions.

3.22.2.9.1 Scuff guards. Tubular or formed scuff guards shall be installed at either side of the catwalk and shall extend the length of the tank. The scuff guards shall be of sufficient height and shall incorporate structural integrity to protect the manhole and other appurtenances mounted on top of the tank in the event of overturn.

3.22.2.10 Boarding ladder. A tubular steel or tubular aluminum ladder shall be securely attached to the rear of the tank to provide access to the catwalk. The ladder shall be mounted in the plane described by the rearmost surface of the bumper and in the center of the tank. The ladder shall be constructed and mounted to provide a rigid structure when scaled by a 220 pound man. A rigid step shall be suspended on the underside of the bumper to facilitate access to the ladder.

3.22.2.10.1 Vertical members. The vertical members of the ladder shall be 18 to 20 inches apart. The vertical members shall extend at least 1 foot above the top surface of the tank and 18 inches horizontally forward of the ladder and then be firmly attached to the tank structure. The portion of the ladder which extends beyond the top portion of the tank shall be removable by one man with common hand tools within 5 minutes for preparing the truck for air transport.

3.22.2.10.2 Safety features. The ladder shall be so designed that it does not interfere with access to the pintle hook, and so that the pintle hook will not be a safety hazard to personnel using the ladder. The ladder assembly shall be inherently safe and convenient for operating personnel.

3.22.2.11 Tank lifting rings . Lifting rings shall be provided and shall be capable of hoisting the empty tank, considering a vertical acceleration of 3g. The lifting rings shall be located at accessible points on the tank to permit cable attachments without damage to any portion of the tank unit.

3.22.3 Pumping compartment. The pumping system shall be housed in a completely enclosed compartment located between the cab and the cargo tank. All pumping system controls, dispensing hoses, meters (mounting spaces) and the suction stub shall be accessible for normal operation from the left or driver's side of the truck.

3.22.3.1 Pumping compartment construction. The pumping compartment shall

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be double constructed and insulated, including the doors. All joints shall be tight. The outer skin shall be a minimum thickness of 18 gage (0.049) sheet steel. The inner skin shall be a minimum thickness of 20 gage (0.035) sheet steel. The insulation shall be at least 1 inch thick, fiberglass, or equivalent and completely enclosed in sheet metal. All panels shall be integrally constructed with reinforcement and permanently welded. The insulating material shall be cemented in place to preclude the possibility of packing and settling. The roof and side panels shall be removable. All doors and hinged sections shall have full length, piano-type hinges with corrosion-resistant hinge pins. The compartment assembly shall incorporate necessary structural integrity to prevent bending, warpage, or other type damage due to vibration and other forces encountered during normal operation. Sharp edges and corners, which could be a hazard to operating personnel, shall not be acceptable. The design and construction of the pumping compartment shall preclude freezing of water-bearing plumbing and components when exposed to an ambient temperature of -20° F for 4 hours with an initial fluid temperature of 130° F.

3.22.3.1.1 Roof. The pumping compartment roof shall be hinged in the center to provide access to and removal of components on either side of the compartment. The roof shall also be removable as an assembly. The roof shall be held in position by hand operated latching devices which will allow access to either side of the compartment or removal of the complete roof by three men without tools in 3 minutes. The design of the roof should allow support for three men standing on top of the compartment without permanent distortion or other damage. The roof panel shall be provided with lifting handles to allow opening of the hinged half sections and removal of the complete panel.

3.22.3.1.2 Doors. Doors shall be provided on the left or driver's side of the compartment. The doors shall be mounted on vertical hinges and shall provide convenient access to the servicing hoses, control panel, and control valves. The doors shall provide an opening substantially the full length of the compartment and at least 36 inches high. One door at least half the length of the compartment and 36 inches high shall be provided on the right side of the compartment. This door shall provide access for inspection and minor maintenance of heater control devices and inspection of the pumping system and its components for leaks, and to allow servicing of components as required. Both the doors on the right and left sides shall be located so a man can enter the compartment through the door openings without difficulty. The doors shall be provided with automatic engaging devices to hold the doors in the open position. The doors and hold open devices shall be so constructed that they are not a hazard to the operator and prevent damage under high winds. All latches hold open assemblies, and mounting hinges shall be mounted in such a manner as to be replaceable with common hand tools.

3.22.3.1.2.1 Door latches. Door latches shall be rigidly constructed and mounted to prevent bending, distortion, or other type failure during long periods of operation. The latches shall be of the quarter-turn release type with a positive-locking means which will not allow the latch to release inadvertantly due to vibration or other forces. Two locking pins per latch shall be provided at the top and bottom of the door. Two locking pins shall engage the side panel structure rather than the opposite door. The latches shall be designed to allow operation by personnel wearing arctic mittens.

3.22.3.1.3 Floor. The pumping compartment floor shall be constructed of at least 3/4 inch marine or exterior grade plywood, smooth on both sides. The ply-

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wood shall be completely covered on both surfaces with galvanized sheet steel adhesively bonded to the plywood. The material bonded to the top surface shall be a minimum thickness of 18 gage. The material bonded to the bottom surface shall be minimum thickness of 20 gage.

3.22.3.1.4 Other equipment housing. The pump, tank-to-pump suction line, tank shutoff valve, line strainer, and tank return line shall be fully enclosed in an insulated housing. Other portions of the pumping system required to be located external of the main pumping compartment shall be included. The insulated housing may be only single constructed with a minimum thickness of 18 gage sheet steel with insulation at least 1 inch thick bonded to the inner surface. The complete housing shall be easily removable as a unit or separate sections. The housing shall be attached with bolts and captive nuts to facilitate removal. Access doors with finger operated latches shall be provided to allow drainage of the pump and removal of the line strainer. The housing shall be so designed that water from the line strainer will not run into the housing or soak the insulation.

3.22.4 Pumping system. The pumping system shall include the tank-to-pump suction line, pump, filter, heater, tank return line, provisions for meters, hose reels, dispensing hoses, nozzles, hose evacuation system, valves, control panel, and associated plumbing and minor components. The pumping system shall be capable of operation for continuous periods of several hours without degrading effects.

3.22.4.1 Tank-to-pump suction line. The tank-to-pump suction line shall be designed to provide adequate flow and smooth operation of the pump under all performance conditions required herein. The tank-to-pump suction line shall incorporate a line strainer, tank shutoff valve, and may incorporate a flexible coupling, if required by the design of the tank and pumping compartment mountings. The tank-to-pump suction line will terminate in the bottom of the tank and not the sump, so that contaminants accumulated in the sump will not enter the pump.

3.22.4.1.1 Line strainer. The tank-to-pump suction line shall incorporate a Y-type line strainer downstream from the suction stub. The strainer shall be a rigidly constructed 40 mesh screen adequately sized to allow a 300 gpm flow at minimum pressure drop. The screen shall be easily removable for cleaning and inspection.

3.22.4.1.2 Tank shutoff valve. The valve shall be of the quarter-turn full-on to full-off type. The valve control shall be easily accessible from the left or driver's side of the truck without requiring the operator to crawl under the truck. The valve shall be located between the tank and flexible coupling, or between the tank and suction stub if the flexible coupling is not furnished.

3.22.4.1.3 Flexible coupling. If required to provide relative motion between the pumping system and the tank, a flexible coupling shall be provided in the tank-to-pump suction line. The coupling shall withstand an internal pressure of up to three times the working pressure or 50 psi whichever is greater, and a vacuum of 0 to 15 inches Hg without leakage or other type of failure.

3.22.4.1.4 Deadman control. When specified (see 6.2) the vehicle shall be

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equipped with a deadman control that stops flow from the tank truck within three seconds and reduces pump discharge pressure in accordance with the NFPA 407. Flow shall be stopped by closing the hose reel valve(s). The deadman control shall consist of an air operated automatic rewind hose with arresting feature for making intermediate position stops while unwinding or rewinding.

3.22.4.2 Pump. The pump shall be of the self-priming, centrifugal type aircraft fuel dispensing pump directly driven by the power takeoff. The pump and pumping system shall provide the performance specified herein when handling demineralized water and fuels conforming to MIL-G-5572 and MIL-T-5624.

3.22.4.2.1 Pump seal failure. The pump shall be so constructed and positioned that in the event of pump seal failure, water will not enter into any gearbox or transmission.

3.22.4.2.2 Pump drain. The pump shall be so designed that it can be completely drained. The drain plug shall be replaced with a drain tube and valve capable of operation by removing the access panel in the insulated housing. The drain tube connected to the pump drain shall be directed to the outside to prevent wetting of the insulation.

3.22.4.3 Suction stub, shutoff valve and dust plug. The tank-to-pump suction line shall incorporate a suction stub upstream of the line strainer. The suction stub shall be a 2 inch size with a manually operated shutoff valve. The suction stub shall be provided with a 2 inch, female, cam-locking quick disconnecting coupling half, with a dust plug and a gasket in accordance with MIL-C-27487. The dust plug shall be retained to the suction stub using 1/16 inch diameter wire rope in accordance with MIL-W-83420. The suction stub shall incorporate a suction breaker capable of restricting vacuum to 5 psig. The suction stub and shutoff valve shall be mounted in the pumping compartment to be easily accessible to an operator standing on the ground at the left or driver's side of the truck. Adequate accessibility shall be provided for connection.

3.22.4.4 Filter. A 300 gpm filter shall be provided. The pumping system shall be so designed that all water dispensed from the hoses, or circulated back into the tank, will have passed through the filter. (The filter shall incorporate twelve MXE-649/E elements in accordance with MIL-F-83428, except that the tests shall be conducted with the elements mounted in the identical housing to be provided by the contractor on the production trucks.)

3.22.4.4.1 Air eliminator. The filter shall incorporate an air eliminator to allow automatic purging of air from the pumping system. The air eliminator shall be vented into the tank.

3.22.4.4.2 Filter drain. The filter shall be provided with a 1 inch manual drain valve capable of draining the complete contents of the filter. The drain line shall extend beneath the floor of the pumping compartment.

3.22.4.4.3 Element support. The upper, or free end, of the filter elements shall be supported by a spider assembly which will hold all the elements in a fixed position. The spider assembly shall be attached to the filter housing.

3.22.4.5 Heater. The heater shall be designed to heat a full tank of demineralized water (2,600 gallons) from plus 32° F to plus 130°F plus or minus 5° within 2 1/2 hours. All electrical components of the heater shall be 24

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equipped with a deadman control that stops flow from the tank truck within three seconds and reduces pump discharge pressure in accordance with the NFPA 407. Flow shall be stopped by closing the hose reel valve(s). The deadman control shall consist of an air operated automatic rewind hose with arresting feature for making intermediate position stops while unwinding or rewinding.

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3.22.4.5.5.5 Fuel filter. A fuel filter shall be provided for removing foreign particles from the fuel. The filter shall be easily removable for inspection and cleaning.

3.22.4.5.5.6 Fuel shutoff valve. A solenoid-operated valve shall be provided to shut off the fuel flow when the cycling limit switch opens, the overheat control switch opens, the master switch opens, the low voltage switch opens, or the automatic purge system is operating. The solenoid valve shall shut off the fuel flow if failure of the ignition system or blower occurs either during start up, or during operation.

3.22.4.5.6 Combustion air blower. The motor driven combustion air blower shall be capable of supplying required air at sufficient pressure to overcome the resistance of the burner and heat exchanger between the air inlet and the heat exchanger gas outlet and shall be capable of purging all combustion gases without the aid of an additional induced draft blower.

3.22.4.5.7 Ignition system. A complete high tension ignition system shall be provided to operate continuously while the fuel shutoff valve is open and give dependable ignition at specified temperature.

3.22.4.5.8 Heater electrical controls. The heater electrical controls shall operate on power from the auxiliary alternator. Electrical switches, etc., required for operating the heater, shall be mounted on a control panel located next to, but separate from, the primary pumping system control panel.

3.22.4.5.8.1 Operating controls. The control panel shall incorporate at least a master electrical switch, heat switch, power-on indicating light, a nocombustion light, and the tank water temperature gage.

3.22.4.5.8.2 Master electrical switch. The master electrical switch shall control power to all electrical components of the heater, except during the combustion chamber purge cycle.

3.22.4.5.8.3 Heat switch. The heat switch shall control the fuel flow to the heater. After the master electrical switch is placed in the on position actuating the blower, the heat switch must be placed in the on position to actuate the fuel solenoid valve and thus produce combustion.

3.22.4.5.8.4 Indicating lights. A green indicating light shall be located over the master electrical switch and shall be illuminated when the switch is in the on position. A red no-combustion light shall be located over the heat switch. The light shall illuminate within 1 minute after actuation of the heat switch, if combustion does not occur, and remain illuminated until the switch is turned off. The no-combustion light shall indicate a malfunction.

3.22.4.5.9 Automatic controls. The heater shall incorporate at least the following automatic controls, and any others, as required to insure safe efficient operation.

3.22.4.5.9.1 High-temperature cutoff switch. A temperature operated switch shall be provided to shut off fuel flow to the heater should the temperature of the water in the heater exceed 205° F plus or minus 5° F. Actuation of this switch shall necessitate manual reset prior to further operation of the heater.

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3.22.4.5.9.2 Tank temperature limit switch. A tank temperature limit switch shall be provided to shut down the heater after the water in the tank has been heated to 130° F plus or minus 5° F. Actutation of the tank temperature switch shall not necessitate manual reset prior to further operation of the heater.

3.22.4.5.9.3 Water temperature cycling switch. In the event the truck is so designed that the water flow rate can be controlled to such a low level that the water temperature in the heater can reach 195° F during operation of the heater, a cycling switch shall be provided. The cycling switch shall shut off the fuel flow at a heater water temperature of 195° F plus or minus 5° F and automatically restart combustion when the water temperature drops 10 to 15° below the cutoff temperature.

3.22.4.5.9.4 Manual reset switch. A manual reset switch or switches shall be provided in the control system. The switch or switches shall require reset after no-combustion malfunction as specified in 3.22.4.5.8.4 and high temperature cutoff as specified in 3.22.4.5.9.1. The manual reset switches shall be located on or adjacent to the heater assembly and shall be accessible from the right side of the truck.

3.22.4.5.9.5 Combustion chamber purge control. The heater control system shall incorporate an automatic combustion chamber purge control. The purge control shall provide sufficient operation after fuel shutoff to limit heater exhaust stack temperature to 400° F or less, even if the shutoff is due to shutoff of the master switch. The control system shall be so designed that fuel will not be supplied to the heater if the purge cycle has not occurred or been completed. In the event the purge cycle does not occur due to 24V dc power failure, the system shall be so designed that the purge cycle must occur prior to the initiation of combustion in the heater.

3.22.4.5.10 Heater electrical power system. The heater electrical power system shall include the 24V, alternator, voltage regulator and controls, and other components as required to provide a safe efficient power supply for operation of the heater.

3.22.4.5.10.1 Alternator. The truck shall incorporate a 24V, dc automotive type alternator. The alternator shall provide all the electrical power for the heater control system. The alternator shall be mounted beneath the pumping compartment and driven by a V-belt from the pump power takeoff driveshaft. The power takeoff driveshaft shall be supported by pillow block mounted on both sides of the alternator drive pulley. The alternator mounting shall be adjustable. The alternator shall be located to be easily accessible for inspection, maintenance, and removal. The alternator and drive mechanism shall be enclosed to protect it from water and road dirt. The enclosure shall allow adequate ventiliation to preclude overheating. The alternator shall incorporate adequate capacity to supply the complete power requirements of the heater and control system without evidence of overheating. Positive means shall be provided to prevent overspeeding the alternator under any condition.

3.22.4.5.10.2 Voltage regulator. An adjustable voltage regulator shall be provided. The regulator shall provide voltage within the required range as specified by the heater manufacturer to insure efficient operation of the heater assembly.

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3.22.4.5.10.3 Circuit breaker. The power system shall incorporate a manual reset circuit breaker to prevent overload of the power system. The circuit breaker and voltage regulator shall be located in a vaporproof box inside the pumping compartment and accessible to an operator standing on the ground. The box shall incorporate hand operated latches for ease of access to components in the box.

3.22.4.5.10.4 Underspeed switch. The heater electrical power system shall include an underspeed switch or low voltage relay to automatically prevent fuel flow to the heater when the alternator speed is insufficient to provide adequate ventilation. The underspeed switch shall be set in accordance with the heater manufacturer's recommendation, and to insure efficient clean burning in the heater.

3.22.4.5.10.5 Electrical system installation. All electrical wiring for the power system and heater control system shall be mounted in protective conduit up to the heater. Components, such as switches and relays subject to moisture and atmospheric degradation, shall be mounted in vaporproof boxes. The entire electrical system shall be separated from the truck electrical system so that the heater system can be removed without disturbing the truck electrical system.

3.22.4.5.11 Water temperature indicator. A tank water temperature indicator shall be installed on the heater control panel. The indicator shall be scaled from 30° F to 150° F in increments of 5°.

3.22.4.5.12 Heater exhaust. The heater exhaust pipe shall be integral with the heater. The exhaust pipe shall vent vertically upward through the pumping compartment roof in the vicinity of the heater. The exhaust pipe shall incorporate a self-closing rain cover and spark arrestor.

3.22.4.5.13 Drain valve. A 1/4 inch easily accessible drain valve shall be provided to drain the heating assembly. The drain line shall extend beneath the floor of the pumping compartment.

3.22.4.5.14 Heater lifting rings. Lifting rings shall be provided and shall be capable of hoisting the empty heater considering a vertical acceleration of 3g. The lifting rings shall be located to enable the heater to be properly positioned on the truck.

3.22.4.5.15 Heater removal. The heater shall be so located that it can be easily removed and a section of pipe inserted between the inlet and outlet connections to utilize the truck for fuel servicing.

3.22.4.6 Tank return line. The pumping system shall incorporate a tank return line to relieve the pump under no-flow conditions during servicing operations, recirculation during tank heating, hose evacuation discharge to the tank, and other functions as required. The return line shall be protected against freezing between the pumping compartment and the tank.

3.22.4.6.1 Flexible coupling. If required to provide relative motion between the pumping system and the tank, a flexible coupling shall be provided in the tank return line and any other line. The flexible coupling shall withstand an internal pressure of up to three times the working pressure or 125 psi, whichever is greater, without leakage or other type failure.

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3.22.4.6.2 Return line check valve. The tank return line, as well as any other lines, with the exception of the tank-to-pump suction line, shall incorporate a check valve between the pumping system and the tank to prevent backflow from the tank when maintenance is being performed in the pumping system. The check valve shall be located between the tank and the flexible coupling or the first detachable joint.

3.22.4.7 Meters. The pumping system shall incorporate space for the installation of meters in accordance with MIL-M-6404 at the inlet to the hose reel. The design of the system shall be such that one section of piping for each meter space can be moved, and the meter installed utilizing standard pipe sections and couplings, and the addition of mounting pads for the meters. When installed, the meters shall be easily readable and within easy reach of an operator standing on the ground in front of the control panel.

3.22.4.8 Hose reel. The pumping system shall incorporate one hose reel conforming to MIL-H-38209 with the exceptions and additions specified herein. The hose reel shall be suitable for servicing the fluids specified in 3.6.2 at flow rates of 0 to 150 gpm. The hose reel shall be provided with a groove-type inlet connection conforming to MIL-C-10387, and 100 feet of 1 1/2 inch ID hose in accordance with MIL-H-26521. Class 2, Type I, male couplings conforming to MIL-C-38404 shall be provided on each end of the hose. The hose reel motor shall be 12V dc, explosion proof and shall operate from the chassis electrical system. A chain drive shall be permitted. A clutch shall not be required if the unwind force requirement can be met without it.

3.22.4.8.1 Hose reel shutoff valve. The valve shall be of the quarter turn, full-on to full-off type. The valve shall be located upstream of the hose reel and shall be easily accessible for the operator standing adjacent to the pumping compartment.

3.22.4.9 Overwing servicing nozzle. One MD-3 overwing servicing nozzle in accordance with MIL-N-87963 shall be furnished on the dispensing hose.

3.22.4.10 Hose evacuation system. The pumping system shall incorporate a hose evacuation system. The evacuation system shall evacuate 90 percent of the water in the fully flooded hose within 1 minute.

3.22.4.11 Valves. The pumping system shall incorporate the necessary valves for safe efficient operation of the pumping system and to meet the performance and functional requirements specified herein. All valves required for operation of the pumping system for water dispensing and hose evacuation shall be of the quarter turn, open and close type or quick opening, gate type. All valves shall be easily operated when the pumping system is under normal operating pressure.

3.22.4.11.1 Manual valves. The system shall incorporate a manually operated quarter turn or quick opening type valve at the inlet to the hose reel as necessary to operate the hose evacuation system. These valves or the valve control handles shall be located adjacent to the pumping system control panel and within easy reach and in clear view of the operator standing on the ground in front of the control panel. The hose reel inlet valve shall allow operation of the pumping system without unreeling the evacuated hose.

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3.22.4.11.2 Drain valves. The pumping system shall incorporate manually operated drain valves as required to insure complete drainage of all portions of the pumping system and all components. The drain valves shall be hand operated and adequately sized to allow drainage in the minimum amount of time. The drain valves shall be easily accessible. The drain lines shall extend beneath the floor of the pumping compartment.

3.22.4.11.3 Automatic valve. The pumping system shall incorporate one or more automatic valves to allow for operation of the bypass system and to perform other functions as required. The valves or valves shall provide for smooth, efficient operation of the pumping system under all operating flow conditions.

3.22.4.11.3.1 Safety provisions. The pumping system shall be so designed that all valves and controls can be adjusted in any position without causing damage to any portion of the truck, with the exception of the hose reel inlet valves, when the hoses are wound on the reels. The bypass line automatic valve(s) shall be provided with a nonadjustable pilot valve which allows the automatic valve to open under all conditions sufficient to relieve the system to a safe pressure slightly above the normal maximum operating pressure.

3.22.4.12 Control panels. The pumping system shall incorporate two control panels on the left of the driver's side of the pumping compartment. The control panels shall be located in the compartment and within easy reach and clearly visible to any operator standing on the ground. The control panels shall be easily removable or easily accessible from the rear for maintenance purposes. The primary pumping system control panel shall contain the auxiliary throttle control, tachometer and combination hourmeter, pressure gages, engine emergency switch, hose reel switches and pumping compartment light switch. The remaining panel shall contain the heater controls.

3.22.4.12.1 Auxiliary throttle. The auxiliary throttle shall provide for control of engine speed during all pumping system operations. The throttle shall allow the operator to control the engine at any desired speed from idle through maximum rated pump rpm as required to conform to the performance requirements specified herein. The throttle shall provide a vernier-type control to allow the operator to select a previously determined operating speed with no difficulty. A mechanism shall be incorporated to allow the throttle to be locked at any desired setting. The mechanism shall be designed such that when the operator strikes the end of the auxiliary throttle with his hand, the lock will be released and the auxiliary throttle will automatically return to the engine idle speed position. An interlock system shall be provided which requires that the parking brake be set before the auxiliary throttle can control the engine speed.

3.22.4.12.2 Tachometer. The control panel shall incorporate a tachometer and hourmeter in combination and conforming to MIL-S-10215 to continuously read engine speed and elapsed time. The tachometer face shall incorporate a red line at the engine speed which should not be exceeded during operation of the pumping system.

3.22.4.12.3 Pressure gages. The control panel shall incorporate two pressure gages to continuously monitor system pressure during operation of the pumping system. One gage shall indicate pump discharge pressure, and the second shall indicate filter differential pressure. The pump discharge pressure gage

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shall have a range of 0 to 150 psi graduated in 2 pound increments. The pressure differential gage shall have a range of 0-100 psi in 1 pound increments. The gages shall incorporate restriction plugs to dampen oscillations and provide stability of the indicating needle. The gages and gage lines shall be so positioned that the fluid handled by the pumping system will drain completely when the pumping system is drained.

3.22.4.12.4 Engine emergency switch. The control panel shall incorporate an engine emergency manual cutoff switch to provide for immediate and complete shutdown of the truck engine from any operating condition without the operation of any other controls. The switch shall be circled with red paint 1 to 2 inches in diameter or otherwise conspicuously marked.

3.22.4.12.5 Hose reel switches. The hose reel switches shall be mounted on the control panel.

3.22.4.13 Piping. The piping layout shall be designed to facilitate maintenance of the pumping system and removal of components with minimum disassembly of piping. Groove type couplings shall be used to the maximum practical extent. Piping sections shall be easily removable for maintenance and repair. Piping connections to major and minor components shall be of the groove type. The piping shall be designed to facilitate draining to the maximum practical extent. The large pipe sections and pipe sections, which could be subjected to excessive movement during operation, shall be firmly mounted or braced. Piping wall thickness shall be a minimum of schedule 40.

3.22.4.13.1 Tubing and conduit. Static pressure and gage sensing tubing lines and electrical conduit shall be installed in groups wherever possible and mounted on blocks which provide for separation of the lines and mounting of multiple lines in one place on any flat surface.

3.22.4.14 Static cable reels. One RMU-25/E reel conforming to MIL-R-83232/2 and one RMU-26/E reel conforming to MIL-R-83232/3 shall be installed inside the pumping compartment in a readily accessible location on the driver's side of the truck. The reels shall be equipped with 90 feet of cable. The location of the reels shall permit easy withdrawal and recovery of the grounding wires. The two reels shall be mounted side by side, and shall be so installed that the resistance between the reel and the chassis frame and tank will not exceed 0.5 OHM.

3.22.4.15 Sampling device. A sampling device shall be provided in the outlet piping to permit sampling the fuel downstream of the filter separator. If possible, the sampling device should be located in a vertical pipe. The sampling device nipple shall extend at least 1/4 inch beyond the inside wall of the pipe from which the sample is obtained. Adequate space shall be provided to permit connection of the Millipore Corporation, Bedford, Massachusetts 01730, Fluid Sampling Kit Catalog Number XX 64 037 00 for solids and water samples. The sampling device shall consist of the following:

a. Nipple, 1/4 inch nominal, schedule 40, aluminum pipe, 2 inches long with one end threaded with 1/4-18 NPT.

b. Valve, ball type, 1/4 inch internal 1/4-18 NPT both ends; aluminum alloy body; highly polished, stainless-steel ball; 300 series stainless steel stem; Teflon seals; cadmium plated malleable iron handle; cadmium

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plated carbon steel handle nut; cadmium plated carbon steel handle stops; quarter turn of handle and valve goes from full open to full close; handle stops limit rotations to 90° plus or minus 5°; pressure loss shall not exceed 1.1 psig at 1.1 gpm of water.

c. Quick disconnect, Snap-Tite Incorporated, Part Number APEC4-4M, manufactured by Snap-Tite Incorporated, Titusville Road, Union City, Pennsylvania 16438; or a coupling of equal construction and characteristics, which will connect to the sampling kit quick disconnect.

d. Dust plug, Snap-Tite Incorporated, part Number AMPE-4, manufactured by Snap-Tite Incorporated; or equal, to fit the previous item, with a 1/16 inch diameter corrosion resistant security wire rope.

e. Teflon tape shall be used on the threads that connect the nipple, valve, and quick disconnect together.

3.22.5 Lights. The truck shall incorporate lights in the pumping compartment and floodlights to facilitate night time servicing.

3.22.5.1 Pumping compartment lights. At least two pumping compartment lights shall be furnished. The lights shall illuminate the entire interior of the compartment and allow observation of all controls and instruments. The lights shall be vaporproof and shall incorporate threaded conduit connections.

3.22.5.2 Night servicing lights. The truck shall be provided with at least two fully adjustable lights to facilitate night servicings. Both lights shall incorporate switches on the adjusting handle. The night servicing lights shall not operate unless the truck driving lights or parking lights are turned on.

3.22.5.2.1 Forward left corner of cab. One light shall be a standard automotive spotlight and mounted in the forward left corner or driver's side of the cab roof. The lights shall incorporate an ANSI Number 4435 lamp. The light shall be so installed that it can be adjusted by the driver, but does not interfere with other normal driving functions or entering or exiting the cab.

3.22.5.2.2 Top left rear corner of tank. The second light shall be located near the top left rear corner of the tank. The light shall be at least 8 feet from the ground, and shall be capable of adjustment by an operator standing on the ground. The light shall be so mounted that it will illuminate the rear and entire left side of the truck. It shall incorporate an ANSI Number 4412 lamp. The light assembly need not be vaporproof.

3.22.6 Fire extinguisher. One 20 pound Halon 1211 fire extinguisher, shall be vertically mounted in a bracket on the outside of the truck on the left side (driver's side). Protection shall be provided from the splashing of mud and water. The fore extinguisher shall be readily accessible to personnel standing on the ground and shall not be mounted behind any pumping compartment door when the door is open.

3.22.7 Wheel chock storage container. A wheel chock storage container 24 inches long, 10 inches deep, and 8 inches high shall be furnished and installed in a space located on the driver's side in an easily accessible location. Bottom of container shall be smooth and have a minimum two 1 inch O.D. drainage

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holes. Container shall be designed for easy removal and storage of wheel chocks.

3.22.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.22.9 Screw threads. Screw threads shall conform to Handbook H28.

3.23 Sound levels.

3.23.1 Exterior sound level. The maximum A-weighted sound level produced by the truck shall not exceed 88 DBA when measured as specified in SAE J366A. An additional 2 DBA allowance over this sound level limit shall be permitted to provide for variations in test site, temperature gradients, test equipment and inherent differences in nominally identical vehicles.

3.23.2 Truck cab interior sound level. The maximum A-weighted sound level produced on the interior of the truck cab shall not exceed 85 DBA when measured in accordance with the procedures of SAE J336, except that the reading described in paragraph 3.7 shall be the maximum A-weighted sound level rather than the maximum levels in each octave band.

3.23.3 Pumping operation sound levels. The maximum A-weighted sound levels produced by the truck during pumping operations shall not exceed 88 DBA at a distance of 15 feet from the geometric center of the truck cab or at the operators position in front of the pumping compartment.

3.24 Piping. The installation of piping, tubing, wiring, etc., shall be accomplished in such a manner as to prevent rubbing and wear due to vibration. The tubing and plumbing shall be assembled to permit adequate flexibility under torsional forces induced during operation. Failure in the piping shall not occur as a result of road operation or torsional forces. Piping, tubing, wiring, etc., which pass through a compartment wall, shall be provided with grommeted holes.

3.25 Finish and protective coatings.

3.25.1 Treatment and painting. The outer surface of the vehicle and interior surface of all compartments except the cab shall be painted in accordance with MIL-STD-1223. Lead and chromate free paint shall be utilized and the paint color shall be Forest Green, Color Number 24052 of FED-STD-595.

3.25.2 Identification. A decal conforming to MIL-M-43719, Type I, Class I, shall be applied to the rear ede of the driver's door. This decal shall contain at least the following data:

a. Contractor rust proofing the vehicle/address.

b. Date of rust proofing.

c. Material(s) used.

3.25.3 Rust proofing areas. Unless otherwise specified, the vehicle shall

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be rustproofed in accordance with the manufacturer's standard practice when specified (see 6.2). The vehicle shall be rust proofed in accordance with MIL-STD-1223. Areas to be protected shall be as described by Table II of MIL-STD-1223 for trucks, as appropriate for this vehicle. Questions pertaining to the appropriate areas should be directed to the procuring activity. All inside areas of the roof and floor panels shall be rust proofed.

3.25.4 Anodizing. The Martin hard coating anodized process shall be given to all aluminum surfaces which are subject to frictional movement relative to adjacent metallic surfaces. All aluminum surfaces, which are not coated in accordance with 3.25.1, and are subject to surface corrosion, shall be anodized in accordance with MIL-A-8625.

3.26 Workmanship. The truck, including all parts and accessories shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts, and tightness of assembly screws and bolts; et cetera.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the Contract or purchse order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products' or supplies' submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the vehicle shall be classified as follows:

a. First Article Testing (see 4.4)

b. Acceptance Testing (see 4.5)

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboatory precision type and shall be calibrated at proper

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intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Water level in cargo tank.

f. Tank water temperature (required for each compartment separated by baffles during all environmental and tank heating tests, control panel tank temperature satisfactory for all other tests).

g. Water temperature at pump inlet.

h. Pressure at pump inlet (in Hg).

i. Pump discharge pressure.

j. Filter outlet pressure.

k. Heater outlet temperature (required for all tank heating tests).

l. Water flow rate (positive displacement meter with an accuracy of plus or minus 0.3 percent over a range of 0 to 300 gpm).

m. Engine speed (rpm).

n. Engine oil temperature.

o. Engine water temperature.

4.3.3 Preparation for tests. The vehicle, equipped with all specified auxiliary equipment, shall be fully serviced; and all equipment shall be prepared for immediate operation.

4.3.4 Lubricants. Lubricants, used in the truck during the tests specified under 4.6, shall be as recommended by the truck chassis manufacturer and the contractor for the mounted equipment.

4.3.5 Test fluid. Unless otherwise specified (see 6.2), the test fluid shall be demineralized water at no more than 10 ppm total solid contaminant.

4.3.6 Test rejection criteria. Throughout all test specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection.

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a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including fuel, coolant, or lubricant, under any operating conditions, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear on tires and in vehicle components.

e. Interference between vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesireable roadability characteristics, including instability in handling during cornering, braking, and while traversing rough terrain.

h. Conditions which present a safety hazard to personnel during operation, service, or maintenance.

i. Overheating of the engine, transmission, or any other vehicle component.

4.4 First Article Testing.

4.4.1 First article test sample. The first article test sample(s) may be a preproduction item(s) or a production item(s) which meets the requirements of this specification. The sample(s) shall be identical to the items procured under the terms of the contract. Approval of the first article sample shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.2 First article tests. The first article tests shall consist of all tests listed under 4.6. If desired, one truck may be subjected to the test specified in 4.6.24.7 and a second truck may be subjected to all other tests.

4.4.3 Test report. Upon completion of the first article test and when required by the contract DD 1423, a first article test report shall be prepared in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report. The following shall be incorporated as a part of the test report.

a. All failures, maintenance performed, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

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c. Summary of the engineering reasoning behind any corrections made, to be made, or proposed to be made, and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements that, if modified, should improve the time.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

4.5 Acceptance testing. Acceptance testing shall be classified as follows:

a. Individual testing (see 4.5.1).

b. Periodic sample testing (see 4.5.2).

4.5.1 Individual tests. Each truck shall be subjected to the following tests as described under 4.6:

a. Examination of product (see 4.6.1).

b. Mechanical operation (see 4.6.2).

c. Static pressure tests (see 4.6.3).

d. Pumping system hydrostatic tests (see 4.6.4)

4.5.2 Periodic sampling plan and tests. One truck shall be selected at random from the first ten manufactured on a production contract, thereafter, one truck shall be selected at random from every fifty or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Pumping system flow test (see 4.6.5)

b. Hose reel tests (see 4.6.10)

c. Road handling test (see 4.6.24.1)

4.5.2.1 Rejection and retest. When a vehicle fails to meet the requirements of this specification, vehicles to be delivered pursuant to the contract may not be accepted until the cause of the failure is determined, corrective action is taken by the contractor and the component that failed is successfully retested to assure compliance with this specification. Tests not involving the failed component may be continued pending the investigation of a test failure.

4.5.2.2 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But, final acceptance of items on hand or items produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.2.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

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4.6 Test methods.

4.6.1 Examination of product. The vehicle shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting, and marking. For the first article, this examination shall be accomplished by use of a checklist which lists each requirement not validated by a test and the results of the examination.

4.6.2 Mechanical operation. A thorough check shall be made of the truck to insure proper functioning of all mechanical parts, accessibility of all controls and levers, interchangeability of removable parts, installation of equipment without fitting, and to effect every degree of safety and performance.

4.6.3 Static pressure test. The truck shall be subjected to an internal static air pressure of 3 psig for not less than 30 minutes. A soap solution shall be sprayed on the exterior of the tank to facilitate detection of leaks. Evidence of leakage shall be cause for rejection.

4.6.4 Pumping system hydrostatic test. All components and lines other than the MD-3 servicing nozzle specified in 3.22.4.9 on the pressure side of the pump shall be subjected to a 30 minute hydrostatic test of 1 1/2 times the maximum total pressure encountered during operation of the unit or 180 psi, whichever is greater. The servicing nozzle shall satisfactorily withstand hydrostatic pressures of 50 and 100 psi for 5 minutes each without leakage, casting failure, or permanent deformation.

4.6.5 Pumping system flow tests. Pumping system flow tests shall be accomplished without any adjustment to the pumping system other than manipulation of controls by the operator in accordance with the operating instructions, except as specifically required herein. All components listed in 3.22.4 shall be installed in the pumping system and shall be adjusted to perform their respective normal function.

4.6.5.1 Flow through overwing nozzle. The overwing servicing hose, with nozzle installed, shall be elevated to recirculate into the manhole of the cargo tank. A flow rate of at least 130 gpm through the hose shall be demonstrated. The condition shall be maintained for 5 minutes. By controlling only the overwing nozzle, flow rates of 0, and approximately 25, 50, 100 and 130 gpm through the hose shall be demonstrated without adjustment or manipulation to the pumping system controls. There shall be no instability in the pumping system operation nor external leakage. The pumping system shall demonstrate rapid flow recovery when the nozzle valve is opened after complete shutoff.

4.6.5.2 Flow limiting. With the maximum throttle setting and the nozzle valves held all the way open, it shall be demonstrated the flow will not exceed 175 gpm through the hose.

4.6.5.3 Flow through suction hose and bypass line. A 50 foot length of 2 inch ID noncollapsible hose shall be attached to the suction stub from a fixed storage tank. A flow rate of at least 100 gpm shall be demonstrated by filling the cargo tank using the system bypass line. This condition shall be demonstrated for at least 5 minutes. The level of water in the storage tank shall be no higher than 6 feet above the truck pump at the beginning of the test.

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4.6.5.4 Hose evacuation system. Water shall be pumped through the servicing hoses to insure the hoses are flooded. The ability of the hose evacuation system to evacuate 90 percent of the water in the flooded hoses within 1 minute shall be demonstrated.

4.6.5.5 Heater operation test. At an ambient temperature no lower than 60° F and an initial tank water temperature of 60° F plus or minus 10°, the truck shall be operated to heat the 2,600 gallons of water in the cargo tank to 130° F plus or minus 5° within 2 1/4 hours. The truck shall be operated and the heater started in accordance with the operating instructions. The truck shall operate unattended until the heating operation is completed and the heater has automatically shut off. In addition to the data required by 4.3.2, the power supply voltage, amperage, and alternator case temperature shall be recorded at 1/2 hour intervals. The truck shall demonstrate the capability to be operated unattended. There shall be no evidence of overheating, power fluctation, or power or control failure during this test.

4.6.6 Closed discharge test. The pump shall be operated at the maximum throttle setting for 30 minutes against a closed discharge. The water temperature in the pump shall not exceed 160° F during this period with an initial cargo tank water temperature of at least 70° F. There shall be no more than 1,000 gallons of water in the cargo tank. The closed discharge condition shall be with the hose reel valves and all other manual valves closed, except the tank shutoff valve.

4.6.7 Dry operation test. The pump and pumping system shall be completely drained. Without introducing liquid into the pumping system, the pump shall be operated at maximum rated rpm for 5 minutes. There shall be no appreciable wear, damage, overheating, nor any other type of failure as a result of this test.

4.6.8 Pumping system malfunction safety test. The dispensing hose shall be completely unreeled from the hose reel and the throttle advanced to the maximum setting. All controls, including valves, shall be randomly manipulated. All adjustable pilots on the bypass valves shall then be adjusted to cause maximum restriction in the bypass valve. All controls shall again be manipulated as specified above. This test shall not produce system pressures which could cause damage, failure or permanent distortion of any part of the pumping system. The nozzles shall be removed from the hoses, and pipe caps installed in the hose ends for this test.

4.6.9 Pumping system endurance test. The pumping system shall be operated at rated flow for at least 100 hours. The system shall be operated continuously for periods of no less than 6 hours throughout the 100 hours endurance test. Each 2 hours of running, the nozzles shall be shut off no less than four times, and then fully opened without changing the truck controls. Compliance with the requirement of 3.21.2.1 shall be demonstrated once. In addition, during each 2 hours of running, the truck engine shall be idled at no-flow for 5 minutes. There shall be no evidence of leakage, wear, part failure or impending failure or any sort of degradation of the truck as a result of this test.

4.6.10 Hose reel tests. The following tests of MIL-H-38209 shall be conducted on the hose reel as installed in the truck:

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a. Mechanical operation.

b. Unwind force.

c. Rewind rate.

d. Rewind force.

4.6.11 Meter installation. Meters conforming to MIL-M-6404 shall be installed upstream of each hose reel to demonstrate the meter installation provisions.

4.6.12 Hydrodynamic loading. The cargo tank shall be filled through the filler cover at a flow rate of at least 600 gpm through a 4 inch diameter pipe. There shall be no evidence of damage, failure, or permanent distortion due to the high flow rate stream. There shall be no excessive splashing due to interfering structure or other characteristics in the design of the tank.

4.6.13 Vent operation test. The vent operation test shall be conducted after the tank has successfully completed the test specified in 4.6.3. One of the two vents, or 50% of the venting capacity, if more than two vents are provided, shall be removed from the tank. All tank openings shall be effectively sealed with the exception of the vent and manometer pressure taps. Water shall be pumped into and out of the tank at 50 gpm. The buildup of pressure and vacuum to plus and minus 0.2 psi, respectively, before the vents open shall be observed. Tank pressures shall be recorded at 15 second intervals, beginning when flow is started into or out of the tank and continuing until the plus and minus pressures in the tank stabilize. Water shall be pumped out of the tank at 130 gpm. The maximum negative pressure shall be recorded and shall not exceed 1 psig. Water shall be pumped into the tank through the system bypass at the maximum possible flow rate up to 130 gpm, but at least 100 gpm. A short suction hose and elevated storage tank shall be used to supply the pump. The positive pressure shall not exceed 2 psig at 130 gpm or proportionately less at lower flow rates.

4.6.13.1 Vent leak test. The venting equipment shall be removed from the tank and installed in a bench test fixture. The fixture shall be subjected to an internal pressure of 5 feet of water and orientated with the axis of the vent horizontal and then vertically downward. The vent leakage rate shall not exceed 40 milliliters per second in either position.

4.6.14 Heater endurance test. The heater, including the heater control system, shall be subjected to an endurance test of at least 120 hours of actual heating operation at rated capacity. Water flow rate through the heater shall be that encountered during operation of the truck. Water flow rate, inlet and outlet temperature, system pressure, voltage, amperage, fuel consumption, and ambient temperature shall be recorded at 1/2 hour intervals. In the event the cycling control is provided, at least 1,000 on-off cycles shall be accomplished automatically by restricting the water flow rate. The heater shall be operated continuously for at least 24 hours. This test shall not produce warpage, deformation, excessive oxidation, control equipment failure or any other evidence of failure, or impending failure, in the heater and control system. The heater and heater controls shall be installed in the first article vehicle for this test in accordance with 4.6.5.5 of this specification and not in a test cell. The test is for the purpose of determining that the heater and controls

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will function correctly after installation in the first article vehicle. The test is unacceptable unless conducted in accordance with specification requirments.

4.6.14.1 Heater efficiency. Based upon the data from 4.6.14, the efficiency of the heater shall be calculated.

4.6.14.2 Heater malfunction safety test. The heater, including the control and power system, shall be subjected to ten externally induced malfunctions for each of the following features. The test shall be conducted on the truck. The heater control system shall automatically shutdown the heater and provide for combustion chamber purge. The exact automatic malfunction procedure, which should occur, shall be explained along with the test results. The heater shall remain off for a period of time to permit observation of the complete combustion chamber purge cycle. All malfunction safety devices shall operate as prescribed without producing any degrading effects to any portion of the truck.

a. Ignition failure while starting heater.

b. Closed or restricted water flow sufficient to actuate the high temperature cutoff switch.

c. Combustion air failure during operation.

d. Alternator underspeed during operation.

e. Power supply failure induced by closing throttle to idle engine.

4.6.15 Flexible coupling hydrostatic tests. All flexible coupling installed in flow lines between the tank and pumping system shall be hydrostatically tested. Each coupling shall be bench tested in a test fixture to simulate the piping on each side of the coupling. The coupling and attachment and clamping arrangement shall be identical to that used on the truck. Flexible couplings on the suction side of the pump shall be tested at 50 psi or three times the maximum working pressure, whichever is greater, and 15 inches Hg vacuum. Flexible couplings in the bypass line or on the pressure side of the pump shall be tested at 150 psi or three times the maximum working pressure, whichever is greater. Flexible couplings on the pressure side of the pump, which are subjected to negative pressures during operation of the unit, shall also be tested at 15 inches Hg vacuum. The couplings shall be subjected to 100 cycles of 0 maximum pressure-0 and 100 cycles of 0-15 inches HG-0, if applicable. There shall be no leaks or evidence of wear, damage, or impending failure as a result of this test. It will not be necessary to subject groove-type pipe couplings to this test.

4.6.16 Tank backflow provisions. The tank shutoff valve shall be closed. The pumping system shall be drained. It shall be demonstrated that the contents of the tank will not flow back into the pumping system under any tank load condition. The test shall be conducted with a tank water level of approximately 1 foot or above the bypass line inlet, whichever is higher, and with the tank approximately filled to capacity.

4.6.17 Structural integrity and part removal tests.

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4.6.17.1 Fender test. It shall be demonstrated that each fender will withstand a static load of 400 pounds placed on the top of the fenders.

4.6.17.2 Catwalk loading. It shall be demonstrated that the catwalk can be loaded to 180 pounds over any 16 square inches or a total load of 600 pounds over any 3 square feet of the catwalk without deformation.

4.6.17.3 Pumping compartment roof loading. It shall be demonstrated that the pumping compartment roof, as installed on the truck, will withstand the weight of three men standing in any position without permanent distortion, damage or evidence of lack of rigidity.

4.6.17.4 Pumping compartment roof removal. It shall be demonstrated that the pumping compartment roof, including the longitudinal center section (see 3.22.3.1.1), can be removed by three men without tools in 3 minutes without any damage to the roof or other portion of the truck.

4.6.17.5 Boarding ladder loading. It shall be demonstrated that a 220 pound man can scale the boarding ladder without evidence of deformation or lack of rigidity of the ladder or associated structure.

4.6.17.6 Boarding ladder removal. It shall be demonstrated that one man with common hand tools can remove the top portion of the boarding ladder, as required for air transport, in less than 5 minutes.

4.6.18 Electromagnetic interference. The truck shall be tested in accordance with MIL-STD-462 for radiated interference in accordance with MIL-STD-461, method RE05.

4.6.18.1 Hose reels tests. Electromagnetic interference tests need not be conducted on the hose reels since the entire truck will be tested as an end item.

4.6.19 Sound level tests.

4.6.19.1 Exterior sound level test. The exterior sound level produced by the truck shall be measured as specified in SAE J366A in order to determine compliance with the requirement specified in 3.23.1.

4.6.19.2 Truck cab interior sound level test. The sound level produced on the interior of the truck cab shall be measured in accordance with SAE J336 to determine compliance with the requirements specified in 3.23.2. The sound level meter used in performing this test shall meet the requirements of ANSI S1.4-1971.

4.6.19.3 Pumping operation sound level tests. The sound levels produced by the truck during pumping operations shall be measured to determine compliance with the requirement specified in 3.23.3. Measurements shall be made at seven positions spaced every 45° on a 15 foot radius circle centered on the geometric center of the truck cab. One of these positions shall be located on the forward longitudinal center line of the truck, and three positions shall be located on each side of this center line. The operator's position shall be located 2 feet from the front of the pumping compartment. At all positions, the microphone shall be located 5 feet above the pavement and oriented so as to obtain the maximum sound reading. Ambient background sound levels, at all

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measurement positions shall be at least 10 DBA below the sound levels produced by the truck. The sound level meter used in performing these tests shall meet the requirements of ANSI S1.4-1971.

4.6.20 Environmental tests. The following environmental tests shall be conducted on the truck in accordance with the specified methods of MIL-STD-810. Unless otherwise specified (see 6.2), an engineering evaluation may be performed in lieu of testing for the tests specified in 4.6.20.3, 4.6.20.4, 4.6.20.5 and 4.6.20.6. The engineering evaluation shall consist of a written detailed analysis of the truck assembly and all its parts and components to satisfactorily accomplish the required environmental conditions.

4.6.20.1 High temperature. The truck shall be subjected to high temperature testing in accordance with Method 502, Procedure I and II. The soak and operating ambient temperature shall be 125° F. The truck shall be filled to capacity with demineralized water at a temperature of at least 75°F at the beginning of the 48 hour soak period. At the end of the 48 hour soak period, the truck engine shall be started. The tests specified in 4.6.2, 4.6.4, 4.6.5.1 and 4.6.5.4 shall be conducted. The truck engine shall then be stopped for 5 minutes and then restarted. The truck shall then be operated at idle speed with the pump engaged for 1/2 hour. The truck engine shall then be stopped for 5 minutes and then restarted. All engine starts shall be accomplished within 5 minutes of the original attempt. In addition to the data required by 4.3.2, the following data shall be recorded during the operational portion of the test at 2 minute intervals:

a. Coolant temperature into engine.

b. Coolant temperature out of engine.

c. Engine oil temperature

d. Automatic transmission oil temperature.

4.6.20.1.1 Satisfactory operation. The truck shall operate satisfactorily to perform its intended functions throughout this entire test. There shall be no leakage, damage, evidence of thermal degradation, or impending failure of any portion of the truck. The engine and transmission oil temperatures shall not exceed the chassis manufacturer's recommended values, which shall be not specified prior to conducting the test. The engine coolant temperature shall exceed that recommended by the manufacturer or 10° below the boiling point rating of the system, whichever is lower. Restarting of the truck engine shall be accomplished without auxiliary cooling or electrical power.

4.6.20.2 Low temperature. The truck shall be subjected to low temperature testing in accordance with Method 502, Procedure I and II. The soak and operating temperature shall be maintained at -40° F and the truck shall not be removed from the chamber until the entire test is completed. After a 48 hour soak period, the truck shall start within 10 minutes of the initial attempt without the addition of auxiliary heating or electrical power. Demineralized water at a temperature of 60° F or less shall be introduced into the cargo tank through the pump and suction stub. Upon filling the tank with 2,600 gallons of water, the heater shall be operated at rated capacity and shall heat the water at a minimum rate of 39.2°F/hr. Upon achieving the 130° F temperature, the truck shall be shut down for a 4 hour period. At the

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end of the 4 hour period, the truck shall be started and shall satisfactorily complete the tests specified in 4.6.2, 4.6.4, 4.6.5.1 and 4.6.5.4. The hose reel shall demonstrate proper operation and adequate power under the low temperature conditions. The truck shall operate satisfactorily to perform its intended functions throughout this entire test. There shall be no leakage, damage, evidence or degradation or impending failure of any portion of the truck.

4.6.20.3 Humidity. The truck shall be subjected to the humidity in accordance with Method 507.2, Procedure I.

4.6.20.4 Fungus. The truck shall be subjected to fungus in accordance with Method 508.3, Procedure I.

4.6.20.5 Salt-fog. The truck shall be subjected to salt-fog in accordance with method 509.2, Procedure I.

4.6.20.6 Dust. The truck shall be subjected to sand and dust in accordance with Method 510.2, Procedured I and II.

4.6.21 Air Transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with 3.6.10. The analysis shall include the tiedowns and all major components and their ability to withstand the accelerations specified in MIL-A-8421. The evaluation shall also include a dimensional analysis for the vehicle while loaded aboard a Type C-130 Aircraft.

4.6.22 Dimensional check. A layout drawing of the truck shall be submitted with all dimensions referenced in the analysis of 4.6.21 measured on the actual vehicle while standing empty on a flat surface.

4.6.23 Weight check. The weight on each wheel of the truck shall be measured with the unit standing on a flat surface, first empty and then fully loaded with 2,600 gallons of demineralized water. The load on each side of the bogie assembly will be acceptable in lieu of the individual wheels of the bogie. The loads shall meet the requirements of 3.8 and 3.22.1.7.

4.6.24 Mobility tests.

4.6.24.1 Road handling test. The truck shall be filled with approximately 2,000 gallons of water and driven not less than 35 miles over paved highway or similar improved roads, and not less than 15 miles over graded gravel or similar unimproved roads. Where practical, the truck speed shall be 45 to 50 mph on improved roads, and 25 to 30 mph on unimproved roads. At least 10 short radius left turns and 5 sudden tops shall be conducted. The wall-to-wall turning circle, both clockwise and counterclockwise, shall be demonstrated. Throughout the roadhandling test, the truck shall be observed as to its trackability, ease of handling, tendency to yaw, tendency to sway, etc. Upon completion of the road handling test, the truck shall be inspected for misalignment, distortion, or other damage sustained as a result of the testing. Undesirable roadability features, and any damage resulting from the test, shall be cause for rejection. A complete description of any adverse handling characteristics or damage sustained shall be recorded on the data sheets.

4.6.24.2 High speed test. The truck shall be loaded to design capacity.

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It shall be demonstrated that the truck can achieve a speed of 50 mph on level paved highways within 2.5 miles from a dead stop. It shall be demonstrated that the truck can achieve a speed of 40mph on level graded gravel road within 1.5 miles from a dead stop.

4.6.24.3 Service brakes. The truck shall be loaded to design capacity. The truck shall be subjected to three emergency stops in rapid succession from 20 mph on clean, dry, smooth, level paved surfaces free from loose material. Each stop shall be accomplished in not more than 1 minute intervals. The truck shall stop within 35 feet from the point the brakes are applied. A time distance recorder shall be used and shall positively identify the point where the brakes were initially applied.

4.6.24.4 Parking brakes. It shall be demonstrated that the parking brakes will hold the fully loaded truck on a 20° ramp in both directions.

4.6.24.5 Ramp clearance. The truck with the cargo tank filled to design capacity shall enter, negotiate and leave a ramp with a slope of 17° in both the forward and reverse directions. There shall be no contact between any part of the truck, except the wheels, with any part of the ramp surface, including the horizontal top and bottom landings. The length of the ramp, measured along the slope, shall be greater than the wheelbase of the truck. The bottom and top of the ramp shall end in horizontal landings which form a sharply defined 17° angle with the ramp.

4.6.24.6 Suspension articulation. The fully loaded truck shall be driven over 8 inch high obstacles. Unless otherwise approved by the procuring activity, the obstacles shall be 8 inch high blocks, one for each track. The blocks shall first be placed side-by-side in each track. These blocks shall be placed next at an angle of 45° across the tracks to simulate the truck negotiating an object at 45° to the path of the truck. The blocks may be sloped to facilitate driving over; however, the blocks will be short enough that when one wheel of the bogie is up 8 inches, the other wheel on that same side will be on the ground. There shall be no interference, part failure, or permanent or objectional distortion of any part of the truck as a result of this test.

4.6.24.7 One thousand six hundred mile test. The truck shall be driven 1,600 miles consisting of four cycles of 400 miles. The first cycle shall be run with the tank half-filled with fuel or water. The second and fourth cycles shall be run with the tank filled to rated capacity. The third cycle shall be run with the tank empty. At the end of the second cycle, the pumping system shall be operated 1/2 hour at rated capacity. At the end of the fourth cycle, the pumping system shall be operated 1/2 hour at rated capacity, and the hydrostatic test of 4.6.4 and the static pressure test of 4.6.3 shall be conducted. During the paved highway portion of each cycle, at least 50 short radius right turns, 50 short radius left turns and 25 sudden stops shall be accomplished. The pumping system, including the filter separator, shall be filled with fuel throughout all four cycles. Each cycle shall consist of the following phases:

a. Highway - 200 miles at an average speed of at least 40 mph and a maximum speed of 50 mph.

b. Gravel road - 100 miles at an average speed of at least 25 mph and a maximum speed of 40 mph.

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c. Belgian block pavement - 100 miles at an average speed of at least 10 mph and a maximum speed of 15 mph.

4.6.24.7.1 Servicing and adjustments. The truck shall be serviced and the brakes adjusted prior to the road test. Periodic servicing with lubricants and preventative maintenance adjustments shall be permitted during the testing at the intervals specified in the operation and service instruction handbook. Tires shall be serviced as necessary. Other servicing, reconditioning or repair shall not be permitted without prior approval of the procuring activity.

4.6.24.7.2 Inspection. Throughout the road endurance test, the truck shall be observed as to its trackability, ease of handling, tendency to yaw, tendency to sway, etc. Upon completion of the road endurance test, the truck shall be inspected for excessive wear, misalignment, distortion, or structural failure of any portion of the truck. If a leak in the pumping system or tank occurs, the complete 1,600 mile road test shall be repeated. If a structural failure occurs, the procuring activity shall determine the extent of retesting. A complete description of any excessive wear, damage, failure, impending failure or roadability deficiencies shall be recorded on the test data sheets.

4.6.25 Reliability and maintainability demonstration.

4.6.25.1 Reliability demonstration. Compliance with the quantitative reliability requirements of 3.6.4 shall be demonstrated by successful completion of all tests specified herein. Apparent failures, that are caused by some unintended factor external to the equipment (e.g. test equipment failure), shall not be counted as reliability failures, if the cause is substantiated to the satisfaction of the procuring activity. Any test resulting in a failure shall be fully documented. Servicing, preventive maintenance, or adjustment authorized by the specification or the procuring activity shall not be counted as failures.

4.6.25.2 Maintainability demonstration. Compliance with the demonstration provisions of the following subparagraphs shall be considered demonstration of compliance with the maintainability and maintenance downtime requirements. Only active maintenance time shall be counted. The capability to perform servicing and maintenance tasks by personnel wearing arctic mittens shall be demonstrated.

4.6.25.2.1 Preventive maintenance downtime. All servicing operations and all preventive and selected schedule maintenance tasks shall be performed. The time, skill level and number of mechanics required shall be recorded. As part of this demonstration, the recommended frequencies of the various tasks, and the times recorded to accomplish the tasks, shall be used to develop an excepted value of preventive maintenance downtime per measure of use, such as hours of operation, miles traveled, or calendar time.

4.6.25.2.2 Scheduled maintenance downtime. Compliance with the corrective maintenance downtime requirments shall be demonstrated by maintenance task performance. The maintenance tasks shall be selected by the procuring activity from an approved list of possible maintenance tasks and estimated performance times, which shall be submitted by the contractor with his first article test procedure. Not less than 25 percent or more than 50 percent of the total number of tasks shall be demonstrated to verify an average maintenance downtime

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within specified limits. Performance times shall be adjusted upward if tasks are demonstrated by personnel possessing higher skills than those specified for task performance. Maintenance tasks performed throughout the test program may be used for maintenance demonstration purposes provided proper documentation is available.

4.6.26 Tools and equipment. Tools and equipment recommended for maintenance shall be used to the maximum practicable extent during testing with emphasis on any special tools or equipment required. Use of special tools and equipment shall be sufficient to determine that they are useful and justified. Instances where special tools or equipment are inadequate shall be recorded in detail.

4.6.27 Safety device tests. The safety devices, such as, only being able to start the engine in neutral/park, inability to simultaneously drive the truck and operate the pump, operation of the auxiliary throttle only with the power takeoff engaged, etc, shall be demonstrated and the results included in the test report.

4.6.28 Rear spring test. The tank shall be half filled with water and shall demonstrate compliance with 3.22.1.19.

4.6.29 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to verify conformance to the requirements of Section 5 and other packaging and marking requirements of the contract.

4.7 Refurbishment. After successful completion of all required testing, each vehicle shall be restored to like new condition.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use. The tank truck shall be used primarily as a demineralized water truck for aircraft, and shall be able to transport the fuels specified herein.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Identification markings required (see 3.15).

c. If lettering and numbering is to be other than specified in 3.15.2.1.

d. If a spare tire, wheel, carrier and tools are required (see 3.22.1.7).

e. Whether or not pintle hooks are to be provided with the production truck (see 3.22.1.10).

f. If deadman control is to be provided with the production truck (see 3.22.4.1.4).

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g. Rustproofing, if other than the manufacturer's standard commercial practice is required (see 3.25.3).

h. If test fluid is to be other than specified in 4.3.5.

i. If an engineering evaluation may be performed in lieu of testing (see 4.6.20).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Pure demineralized water. Pure demineralized water is defined as water meeting the following requirements:

a. Total solids must not exceed 10 ppm.

b. Acidity/alkalinity (PH) shall be from 6.0 to 7.5.

6.3.2 Mean time between failures (MTBF). MTBF, for reliability purposes, is defined as the average (arithmetic mean) operating time between failures that prevents satisfactory performance of primary functions. Note that MTBF, demonstrated at a stated confidence level, will be less than a point estimate derived by dividing the accumulated operating time by the number of failures. Also, the predicted MTBF should be greater than the demonstrated MTBF.

6.3.2.1 Failure. For reliability purposes, a failure is defined as any malfunction that could prevent satisfactory, within specified limits, and safe operation in the primary functions of loading, traveling to and from a delivery site, and pumping fluid.

6.3.3 Self-priming centrifugal pump. A self-priming centrifugal pump is defined as one designed to have the ability to prime itself automatically, after being initially filled, when operating under a suction lift, and to free itself of entrained gas without losing its prime and to continue normal pumping without attention.

6.3.4 Hazardous vapor area. The hazardous vapor area is defined as the entire rear portion of the truck from the back of the cab.

6.3.5 Specific gravity t/T. Refers to the density at a temperature t, divided by the density of pure water at temperature T.

6.3.6 Highway. Highway is defined as a high speed paved road which is essentially a level road that permits the operation of most military vehicles at maximum speed.

6.3.7 Gravel road. A gravel road is defined as a compacted gravel surface which is maintained by grading and containing left and right curves.

6.3.8 Belgian block. Paved unevenly layed granite blocks forming an undulating surface. It duplicates a rough, cobblestone road.

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6.4 Subject term (key word) listing.

a. Truck, Tank
b. Pumping System
c. Engine, Diesel
d. Demineralized Water
e. Heater, Tank
f. Specification, Military

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 84
Agent: Air Force - 99
Project Number:
2320-F293

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-26510E	TRUCK, TANK A/S32A-2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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PREVIOUS EDITION IS OBSOLETE.

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